

Farm Buddy PRD

1. Product Overview

Farm Buddy is a smart agriculture platform designed to help farmers monitor crops, maintain farm records, and provide supply chain traceability using AI and blockchain.

2. Problem Statement

Farmers face crop diseases, lack of proper records, no supply chain transparency, and consumers cannot verify product authenticity.

3. Objectives

- AI-based crop guidance
- Digital farm records
- Disease detection
- Supply chain traceability
- Consumer trust

4. Target Users

Primary: Farmers

Secondary: Buyers, Consumers, Distributors

5. Core Features

- Farmer Dashboard
- Crop Health Monitoring
- AI Chatbot (Gemini API)
- Activity Logging
- QR Code Generation
- Supply Chain Traceability

6. Technology Stack

Frontend: React.js, Tailwind CSS

Backend: Node.js, Express.js

Database: MySQL

AI: Gemini API

Blockchain: Solidity (Future Phase)

7. System Flow

1. Farmer Actions: Planting / Irrigation / Fertilization / Harvest / Lab Test etc.
2. Data Ingestion Layer: Web/Mobile App, Farm Buddy Voice App, IoT Sensors.

3. Backend API (Node.js/Express): Validates data, assigns Batch ID and timestamps.
4. MySQL Database: Stores complete batch, event, operator, and sensor details.
5. Hashing Engine: Generates SHA-256 hash of event data (Chain of Custody).
6. File Storage (IPFS): Uploads lab reports/images, receives IPFS CID/URI.
7. Smart Contract: recordFarmEvent(batchId, ipfsHash, dataHash, timestamp) on-chain.
8. Blockchain Ledger: Stores Batch ID, Data Hash, IPFS CID, and Transaction ID.
9. Batch Ready for Market: Quality checked and approved.
10. QR Code Generation: Generates unique QR code with Batch ID and tracking URL.
11. QR Code Printed on Packaging.
12. Consumer Scans QR Code: Scans using any smartphone.
13. Consumer Portal (Web App): Queries Blockchain and Backend API for Batch ID.
14. Display Traceability Timeline: Displays crop events, sensors, and IPFS reports.
15. Verification Process: Recalculates hash and fetches blockchain hash.
16. Hash Match Verification: Compares recalculated hash with on-chain hash.
- 17A. Verified: Hashes match. Data is authentic and untampered.
- 17B. Tampered: Hashes do not match. Data is flagged as modified or corrupted.

8. Functional Requirements

FR1: User Registration/Login

FR2: Add Farm Details

FR3: Upload Crop Images

FR4: AI Analysis

FR5: Store Activity Logs

FR6: Generate QR Codes

9. Non-Functional Requirements

Fast response, secure storage, scalability, simple UI, multilingual support

10. Future Scope

IoT integration, weather forecasting, smart irrigation, blockchain verification, market prediction

Conclusion

Farm Buddy aims to modernize agriculture through AI-powered assistance and transparent product traceability.